

Aaron Parker

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with 6+ years of experience building and maintaining production applications across the full stack. Strong backend, database, and infrastructure depth (C#, Java, Go, TypeScript, Python, PostgreSQL, SQL Server, Kubernetes) with end-to-end ownership from design through deployment. Experienced in translating complex domain requirements into clean, scalable systems for cross-functional research and operations teams, with a track record of mentoring junior and student developers.

TECHNICAL SKILLS

Languages: C#, Java, C++, Go, TypeScript (Node.js), Python, Bash

Microsoft Stack: C#, .NET, SQL Server, Azure DevOps Pipelines, Azure cloud infrastructure

Backend & APIs: Express, Fastify, GraphQL, Flask, REST API design

Databases: SQL Server, PostgreSQL, MySQL, MongoDB, SQLite, Neo4j, Prisma

Frontend: React, TypeScript, URQL/GraphQL, Redux, responsive UI architecture, real-time data handling (WebSocket/WebRTC)

Infrastructure & DevOps: Kubernetes (Deployments, Services, Ingress, PV/PVC, RBAC), Docker, CI/CD automation, GitHub Actions, Git

Specialized: Unity, real-time video/audio pipelines, Vuzix Blade, and HoloLens 2 XR development

CERTIFICATIONS

- Certified Kubernetes Application Developer (CKAD), CNCF

EXPERIENCE

Software Engineer (Full-Stack)

2022 – Present

Deaf HUB, Rochester Institute of Technology

- Owned full feature lifecycle from requirements gathering through design, implementation, deployment, and post-launch maintenance for production web platforms running in multi-tenant Kubernetes environments.
- Built and maintained REST and GraphQL APIs using Node.js, Prisma, and Go, with data persistence across PostgreSQL and cloud databases supporting research and production workloads.
- Engineered real-time captioning and interpreter-streaming features synchronizing live video and caption data over WebSocket connections, serving Deaf and Hard-of-Hearing users across multiple research deployments.
- Translated complex domain and operational requirements from cross-functional research teams into production-ready software systems, including live captioning, interpreter streaming, and XR caption delivery on Vuzix Blade and HoloLens 2.
- Provided technical guidance and code review for junior engineers and student developers on backend architecture, containerization, and deployment practices.
- Implemented CI/CD pipelines enabling rapid feature iteration and reducing deployment cycle time.
- Architected scalable containerized services deployed to Kubernetes with automated monitoring and resource management.

DevOps Engineer

2020 – 2022

Center on Culture & Language, Rochester Institute of Technology

- Developed backend services and deployment pipelines for production research applications running in Kubernetes.
- Containerized several research applications, replacing brittle manual deployment steps that had been a recurring source of bugs.
- Added Prometheus metrics to backend services so we could see what was happening in production, not just guess.
- Wrote and shipped an Android app for the Vuzix Blade smart glasses and worked through the HoloLens 2's debugging quirks for our XR accessibility research.
- Managed the Azure environment and our Linux web servers; mentored several NTID student developers on Docker, Kubernetes basics, and how to deploy without breaking things.

Mobile Application Developer

2018 – 2020

Center on Access Technology, Rochester Institute of Technology

- Team leads for the Teach 2 Connect application, published across all platforms in time for RIT teacher orientation.
- Developed cross-platform mobile applications using C# and Xamarin with cloud-backed data storage and real-time synchronization.
- Designed API integrations, database schemas, authentication flows, and mobile-to-cloud data pipelines.

SELECTED PROJECTS

Access on Demand: Real-Time XR Captioning & Interpreter Livestream

- Built a Unity application that connects to a web server to synchronously render live captions and a real-time interpreter video feed inside a 3D environment, delivering remote access services on Smart Glasses (Vuzix Blade) and HoloLens 2.
- Presented at the Association of Medical Professionals with Hearing Losses (AMPHL): "Increasing Accessibility in Health Care with Smart Glasses," including a live demonstration (2022).

NTID 50th Reunion Scheduling App

- Built a multi-platform event app with asynchronous, real-time schedule updates; received positive feedback from over 3,000 users.

Dyer Art Center Exhibition Support

- Improved back-end services for a cross-platform app letting visitors learn about artworks via numeric input, QR code, or NFC; recognized by the RIT president during the President's Alumni Ball address.

Deaf Refugee Conference App

- Cross-platform conference app for the Reach Out! Supporting the Deaf Refugees conference in Rochester, NY; fetched schedule data from a backend and cached it for full offline use. Published to Google Play.

EDUCATION

B.S. Web & Mobile Computing, Rochester Institute of Technology, 2021

A.A.S. Mobile Application Development, National Technical Institute for the Deaf, 2019